

Final Project Report: The Impact of Age on NBA Player Performance

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SAS Programming and Data Management

STA 6033

May 14, 2026

1. Introduction

One of the greatest sources of entertainment for Americans for decades has been the National Basketball Association (NBA), a place where some of the world's greatest athletes can showcase their talents. Beginning in New York City over 70 years ago, Walter Brown observed that tens of thousands of people were coming to watch hockey games, giving him the inspiration he needed to promote basketball games in these same arenas (NBA Careers, n.d.). With over 77 years of history, we've seen some of the greatest athletes on the planet come through the league, some of whom you will find defy the odds of age being a factor in their performance.

For this project, I have chosen three datasets consisting of data from over 400 NBA players and 53 performance statistics to analyze these NBA players' careers. This analysis will determine **whether age impacts a player's overall performance on the court** from a data-driven perspective.

1.1 Data Limitations

The various age cohorts within the datasets range anywhere from 1-46 players. To ensure that sufficient data was present, age cohorts with less than 5 players were filtered out. The analysis is performed on NBA players ranging from ages 19-35.

2. Data Processing

To load the data from Microsoft Excel into Statistical Analysis Software (SAS) the 'Proc Import' step was used combined with the Database Management System command (DBMS). The 'DBMS=csv' command informed SAS studio that it would be connecting to a Comma-Separated Value file (CSV). Next, for the purposes of conjoining the datasets, the character variable 'ID' (player ID) was identified as the primary key for merging the three datasets together.

2.1 Relational Data Integration

SAS Studio requires data to be sorted by the primary key prior to merging. To ensure a clean merge, the 'PROC SORT' step was used to sort the datasets by their primary key show below. The 'Data' step was then used to both merge the three datasets and standardize the formatting of the variables. The result was a dataset consisting of 53 variables and 560 rows – far too many variables for a focused analysis. The excessive number of variables is addressed in the data cleaning section.

Filter: (none)

Total rows: 839 Total columns: 3

	Salary	Player	ID
1	\$2,582,160	Precious Achiuwa	achiupr01
2	\$29,592,695	Steven Adams	adamast01
3	\$5,115,492	Bam Adebayo	adebaba01
4	\$16,750,000	LaMarcus Aldridge	aldrila01
5	\$878,340	LaMarcus Aldridge	aldrila01
6	\$16,750,000	LaMarcus Aldridge	aldrila01
7	\$878,340	LaMarcus Aldridge	aldrila01
8	\$16,750,000	LaMarcus Aldridge	aldrila01
9	\$878,340	LaMarcus Aldridge	aldrila01
10	\$16,750,000	LaMarcus Aldridge	aldrila01
11	\$878,340	LaMarcus Aldridge	aldrila01
12	\$3,113,160	Nickel Alexander-Walker	alexani01
13	\$2,545,320	Grayson Allen	allengr01
14	\$3,909,902	Jamett Allen	allenja01
15	\$3,909,902	Jamett Allen	allenja01
16	\$3,909,902	Jamett Allen	allenja01
17	\$3,909,902	Jamett Allen	allenja01
18	\$9,720,900	Al-Farouq Aminu	aminual01
19	\$9,505,100	Kyle Anderson	anderky01
20	\$27,528,088	Giannis Antetokounmpo	antetgi01
21	\$1,701,593	Thanasis Antetokounmpo	antetth01
22	\$2,564,753	Carmelo Anthony	anthoca01
23	\$3,285,120	Cole Anthony	anthoco01
24	\$3,872,215	OG Anunoby	anunoog01
25	\$3,000,000	Ryan Arcidiacono	arcidry01
26	\$12,800,000	Trevor Ariza	arizar01
27	\$6,666,667	D.J. Augustin	augusdj01
28	\$6,666,667	D.J. Augustin	augusdj01
29	\$6,666,667	D.J. Augustin	augusdj01
30	\$6,666,667	D.J. Augustin	augusdj01
31	\$6,666,667	D.J. Augustin	augusdj01
32	\$6,666,667	D.J. Augustin	augusdj01
33	\$6,666,667	D.J. Augustin	augusdj01

3. Data Structure

To begin this analysis, information from three datasets consisting of salary data, advanced player statistics data, and game data was used for this project. To visualize the metadata from each of these datasets within SAS programming studio the proc contents procedure was used to show summary statistics shown below. The metadata shows the first dataset containing 3-character variables, the second dataset 6-character variables and 23 numerical variables, and the third dataset containing 31-character variables and 1 numeric variable.

Salary Data

Data Set Name	NBA.SALARY	Observations	839
Member Type	DATA	Variables	3

Advanced Stats data

Data Set Name	NBA.ADVANCED_STATS	Observations	456
Member Type	DATA	Variables	29

Game data

Data Set Name	NBA_GAME_STATS	Observations	461
Member Type	DATA	Variables	32

Salary data

Advanced Stats data

Game data

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
3	ID	Char	11	\$11.	\$11.
2	Player	Char	26	\$26.	\$26.
1	Salary	Char	13	\$13.	\$13.

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
15	AST.	Num	8	BEST12.	BEST32.
2	Age	Num	8	BEST12.	BEST32.
17	BLK.	Num	8	BEST12.	BEST32.
26	BPM	Num	8	BEST12.	BEST32.
25	DBPM	Num	8	BEST12.	BEST32.
13	DRB.	Num	8	BEST12.	BEST32.
21	DWS	Num	8	BEST12.	BEST32.
11	FTr	Num	8	BEST12.	BEST32.
6	G	Num	8	BEST12.	BEST32.
29	ID	Char	11	\$11.	\$11.
4	Lg	Char	5	\$5.	\$5.
7	MP	Num	8	BEST12.	BEST32.
24	OBPM	Num	8	BEST12.	BEST32.
12	ORB.	Num	8	BEST12.	BEST32.
20	OVS	Num	8	BEST12.	BEST32.
8	PER	Num	8	BEST12.	BEST32.
5	Pos	Char	7	\$7.	\$7.
16	STL.	Num	8	BEST12.	BEST32.
1	Season	Char	9	\$9.	\$9.
18	TOV.	Num	8	BEST12.	BEST32.
14	TRB.	Num	8	BEST12.	BEST32.
9	TS.	Num	8	BEST12.	BEST32.
3	Tm	Char	5	\$5.	\$5.
19	USG.	Num	8	BEST12.	BEST32.
27	VORP	Num	8	BEST12.	BEST32.
22	WS	Num	8	BEST12.	BEST32.
23	WS.48	Num	8	BEST12.	BEST32.
10	X3PAr	Num	8	BEST12.	BEST32.
28	player	Char	26	\$26.	\$26.

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
25	AST	Char	43	\$43.	\$43.
2	Age	Num	8	BEST12.	BEST32.
27	BLK	Char	43	\$43.	\$43.
23	DRB	Char	43	\$43.	\$43.
9	FG	Char	43	\$43.	\$43.
11	FG.	Char	43	\$43.	\$43.
10	FGA	Char	43	\$43.	\$43.
19	FT	Char	43	\$43.	\$43.
21	FT.	Char	43	\$43.	\$43.
20	FTA	Char	43	\$43.	\$43.
6	G	Char	43	\$43.	\$43.
7	GS	Char	43	\$43.	\$43.
32	ID	Char	11	\$11.	\$11.
4	Lg	Char	43	\$43.	\$43.
8	MP	Char	43	\$43.	\$43.
22	ORB	Char	43	\$43.	\$43.
29	PF	Char	43	\$43.	\$43.
30	PTS	Char	43	\$43.	\$43.
5	Pos	Char	43	\$43.	\$43.
26	STL	Char	43	\$43.	\$43.
1	Season	Char	9	\$9.	\$9.
28	TOV	Char	43	\$43.	\$43.
24	TRB	Char	43	\$43.	\$43.
3	Tm	Char	43	\$43.	\$43.
15	X2P	Char	43	\$43.	\$43.
17	X2P.	Char	43	\$43.	\$43.
16	X2PA	Char	43	\$43.	\$43.
12	X3P	Char	43	\$43.	\$43.
14	X3P.	Char	43	\$43.	\$43.
13	X3PA	Char	43	\$43.	\$43.
18	eFG.	Char	43	\$43.	\$43.
31	player	Char	26	\$26.	\$26.

3.2 Data Type Standardization

To ensure data standardization among the newly merged dataset, the columns were renamed and filtered using 'RENAME' statement. Several of the columns had inconsistencies in their formatting for example, the advanced stats dataset marked the AST and BLK variables as numeric while the game dataset marked these as character variables. Additionally, columns such as the 'TS.' column required the 'n.' name literal to ensure that this variable was treated as an actual variable and not anything else due to the trailing period. The remaining columns were renamed to contain non-acronym names shown below. Lastly, the BEST32. format was used to format the character variables that needed to be listed as numeric, this format is the SAS default for displaying numeric variables in a tidy fashion.

```

rename
  player = player_name
  Season = NBA_season
  Age = player_age
  Pos = Position
  Tm = Team
  MP = minutes_played
  G = games_played
  PER = player_efficiency_rating
  BPM = box_plus_minus
  VORP = value_over_replacement_player
  WS = win_shares
  PTS = points_scored
  AST = Assists
  TRB = total_rebounds
  STL = Steals
  BLK = Blocks
  TOV = Turnovers
  'TS.'n = true_shooting_pct;

```

3.3 Data Cleaning and Outlier Management

To manage outliers, the `PROC UNIVARIATE` step was used to analyze the total games that the NBA players participated in by age group. The result was a quantiles table that was used to reveal outliers shown below. Outliers in the 1% category and 0% Min category revealed that only one data point lived in this section, identifying and managing these outliers prevented the results of this analysis from being majorly skewed. Additionally, the `G >= 10` filter helped to remove several major outliers of injured players that had entries reading “Did Not Play”.

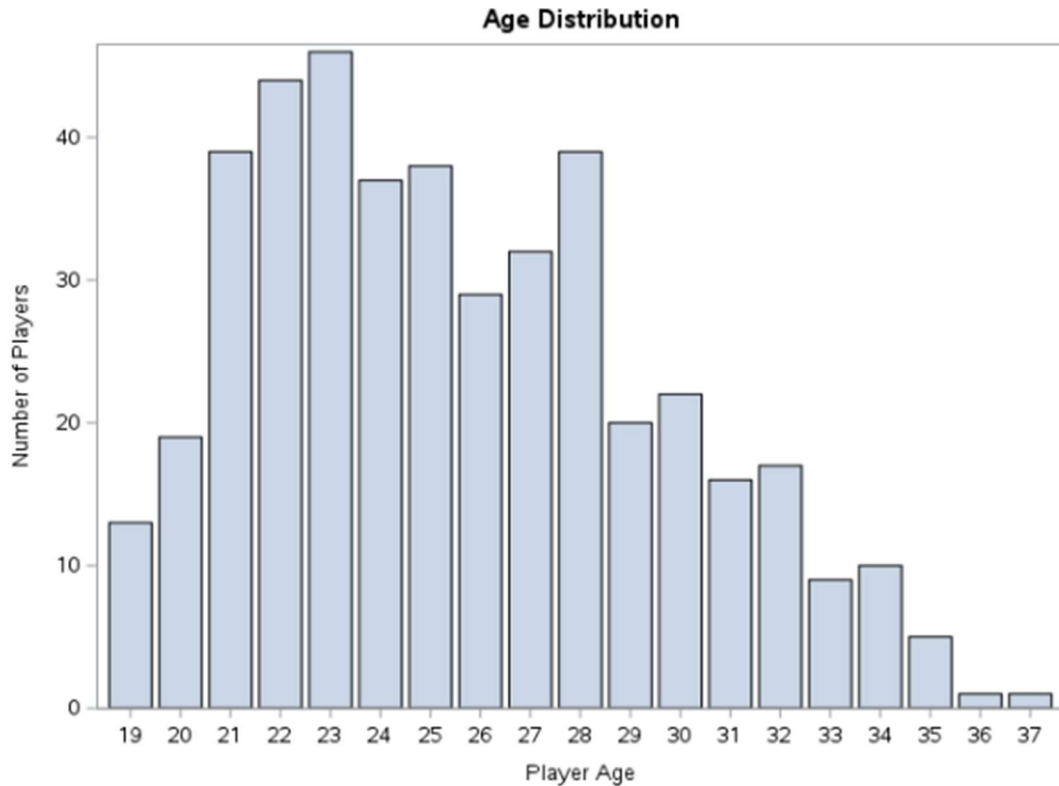
Before:		After:	
Quantiles (Definition 5)		Quantiles (Definition 5)	
Level	Quantile	Level	Quantile
100% Max	72	100% Max	72
99%	72	99%	72
95%	69	95%	69
90%	66	90%	67
75% Q3	57	75% Q3	58
50% Median	35	50% Median	37
25% Q1	22	25% Q1	23
10%	16	10%	18
5%	11	5%	15
1%	1	1%	11
0% Min	1	0% Min	10

After performing an audit on the merged dataset, unnecessary variables and fields were removed using the `KEEP` and `WHERE` command. This audit included filtering out players who did not log active minutes during the 2020-21 season (often marked as "Did Not Play"). Removing these inactive entries prevented the skewing of summary statistics and performance averages. The

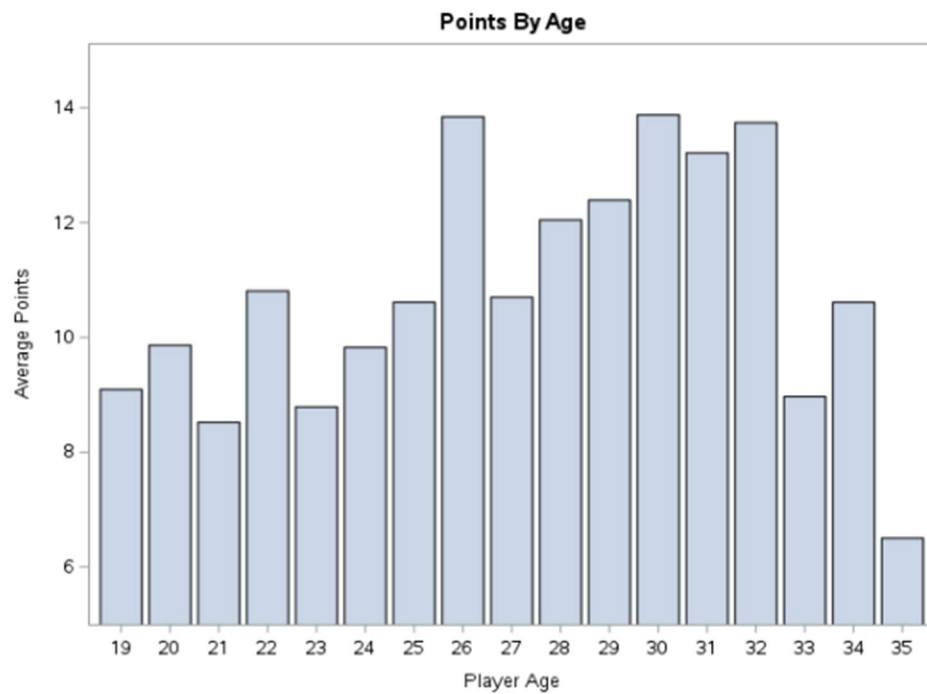
'KEEP' statement reduced the number of variables in the dataset from 53 to 19 and the 'WHERE' statement was used to filter out NULL age values and remove players who did not play more than 10 games. The result of the refined dataset contained 19 columns and 611 rows.

3.4 Exploratory Data Analysis

Next, to begin analyzing the variables individually the age categories needed to be addressed. The distribution chart below reveals the skewed number of players within each age category.

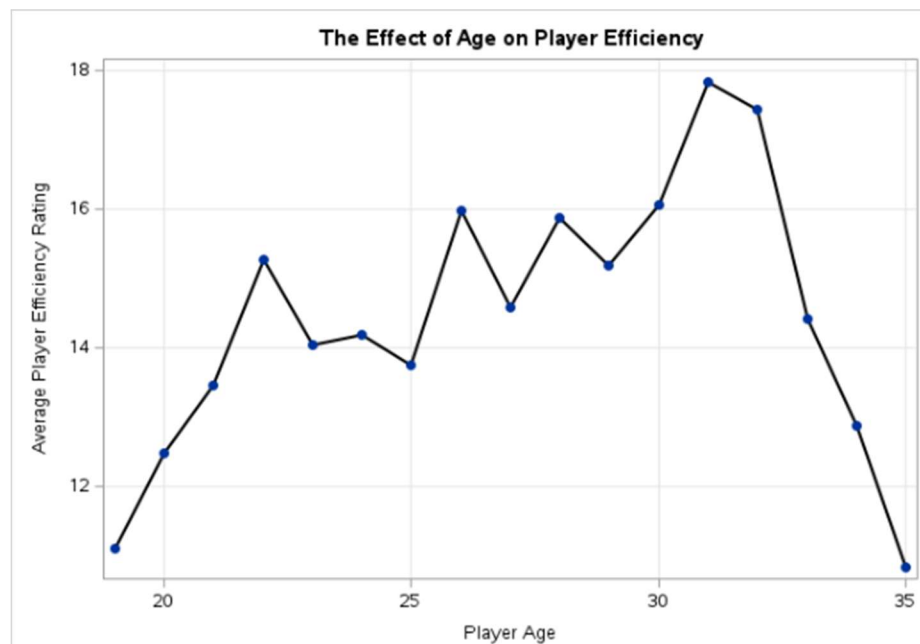


Findings: The frequency chart below shows the variation of age within the data set before data cleaning, showing older players appearing less frequently with a major spike in ages 21-25 in datapoints. This graph also revealed outliers such as ages 36 and 37 that appeared in the dataset less than 5 times. To resolve these outliers, players that had less than 5 data entries were removed to ensure that the results were balanced with sufficient data for each age category.



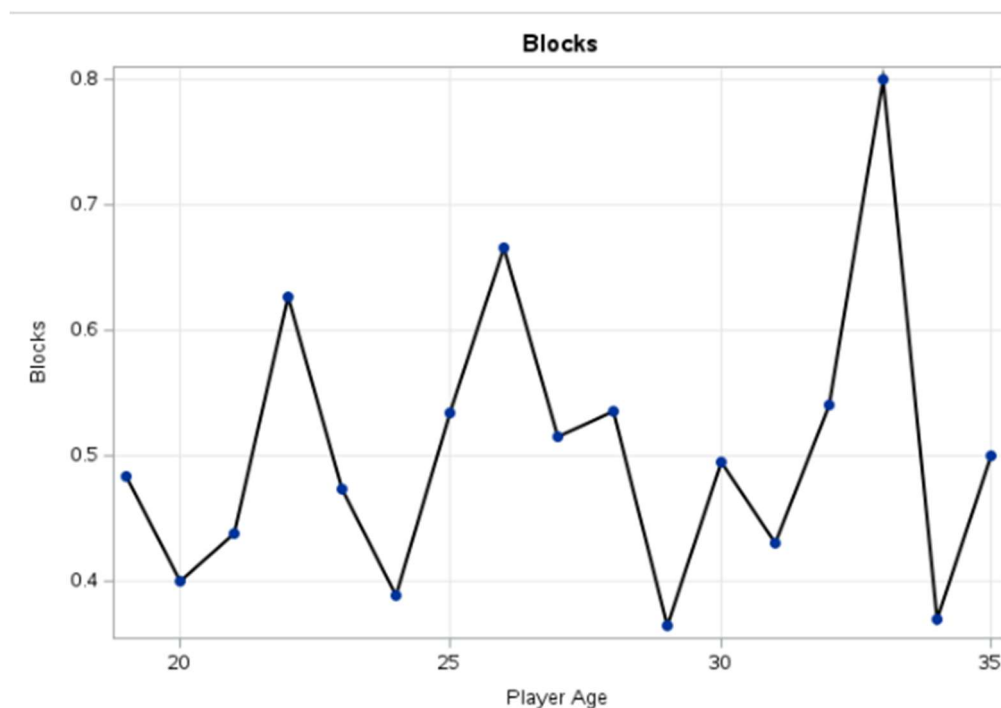
Findings: The points distribution by age graph revealed a spike in average points between age 25 that was relatively consistent through age 32.

4.1 The Effect of Age on Player Efficiency Rating (PER)



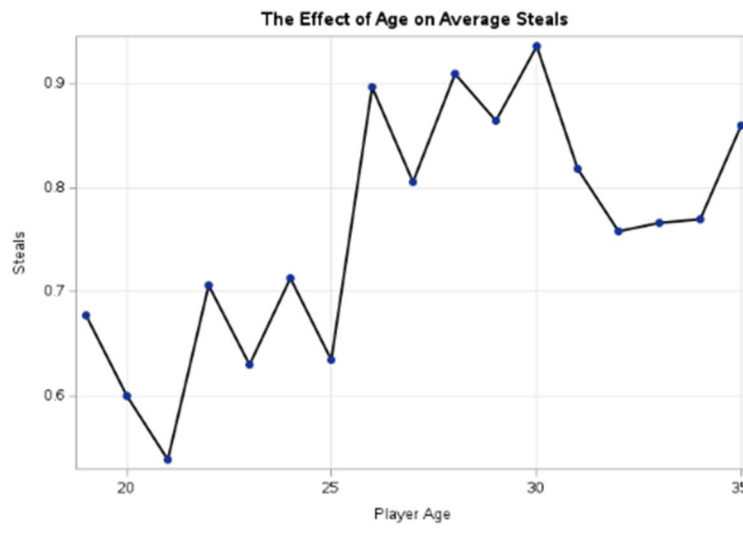
Findings: The next variable that was evaluated was Player Efficiency Rating (PER). PER is defined as a measure of an NBA players per minute productivity (Hollinger, 2011). **The results showed that players that were 22, 29, 30, 31, and 32 had a higher efficiency rating ranging anywhere from 16.1 - 17.5.** The noticeable performance spike in players in their late twenties and early thirties is likely caused by players hitting what is often known as a 'sports prime.' Following the sports prime, there is a visible drop in PER amongst players who were above the age of 32.

4.2 The Effect of Age on average blocks per game



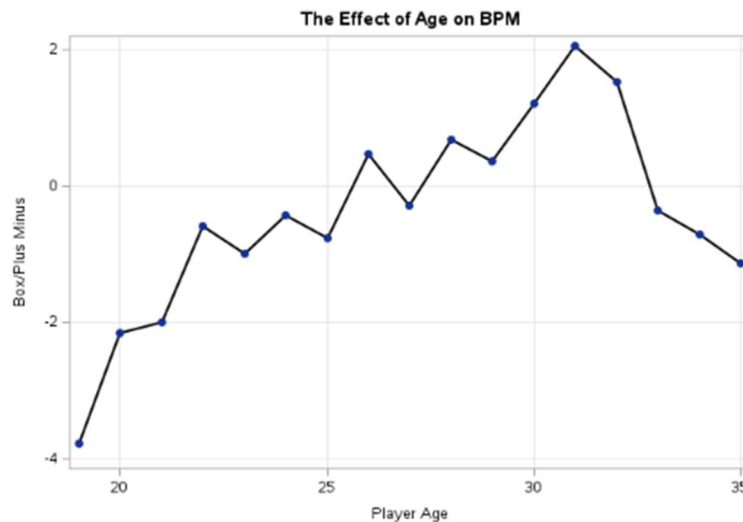
Findings: Blocks were distributed more sporadically throughout the data with a large spike between 32-33. This variable was useful for showing the more mature players average more blocks.

4.3 The Effect of Age on Steals



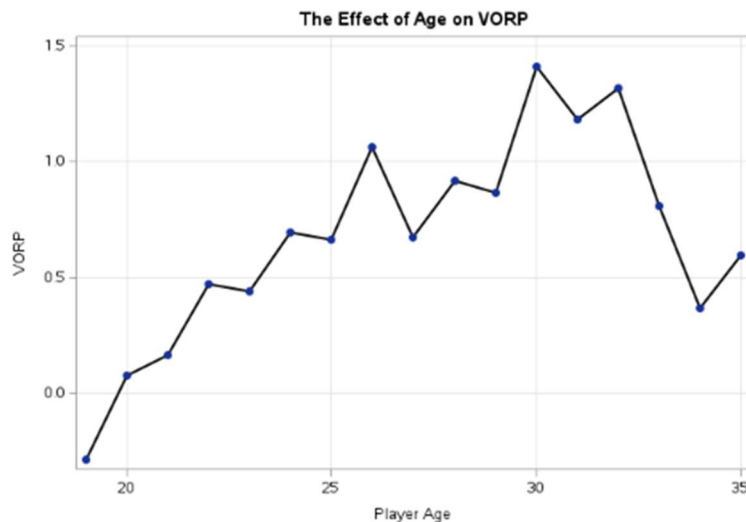
Findings: The Steals graph shows evidence that supports the 'Athletic Prime' hypothesis. Players are averaging more steals per game between age 26 – 31.

4.4 The Effect of Age on Box Plus Minus



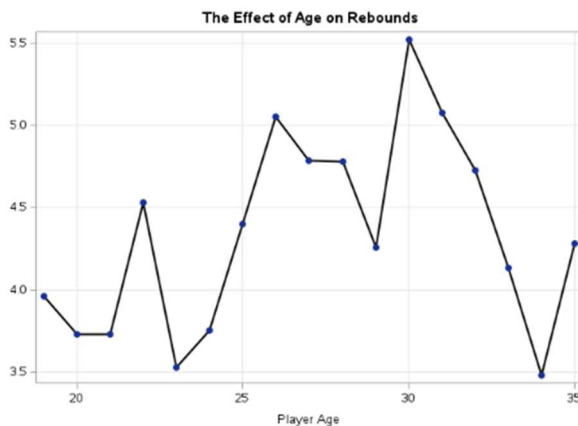
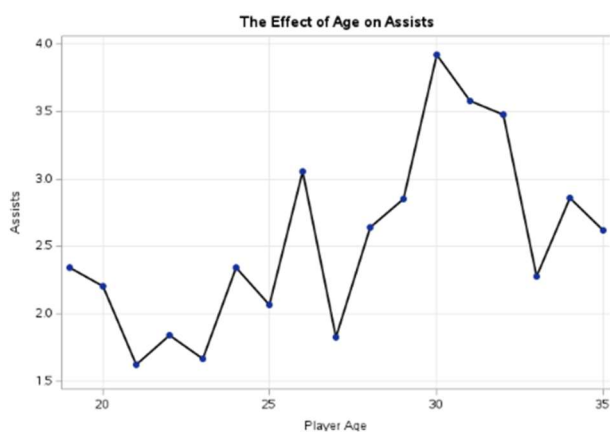
Findings: A BPM score is a metric in basketball used to determine a player's contribution in points to the team while they are on the floor per 100 possessions. The higher the score the better. This graph shows clear evidence in favor of the Sport Prime hypothesis that players perform better between ages 26-32.

4.5 The Effect of Age on VORP



Findings: The VORP is metric used in basketball to determine the value over a replacement player a player provides. The VORP graph shows strong evidence for VORP being a measure of the Sports Prime in players with a spike between ages 24-32.

Additional Graphs:



These two graphs show the as players age into their late twenties and early thirties then tend to have higher rebounds and assists. Based on the evidence in this study, higher rebounds and assists tend to be attributed to more mature players.

5. Summary of Findings

- Strong evidence suggests the NBA players perform at their highest level between the ages of 26 – 32. The data showed that **players in the 26 – 32 age brackets average higher Points, PER, VORP, Steals, and BPM.**

- Strong evidence suggests that **mature players between ages 30 – 32 have the highest BPM, Assists, Rebounds, Steals, and VORP.**
- The data suggests that players experience their lowest recorded performances in the Points and PER categories after age 34.
- In categories such as PER, Rebounds, Assists, Steals, Blocks, and VORP players experience the highest performance of their prime between ages 30 and 31.

6. Conclusion

In conclusion, the results of this study reveal that age does in fact impact the performance of any NBA player. The data strongly suggests the player experience their high performing moments across several different metrics between the ages of 26 - 32. The data also reveals that mature players between age 30 and 31 experience high-performance averages in several categories.

References

NBA Careers. (n.d.). History. NBA Careers.

Hollinger, J. (2011, August 8). *What is PER?* ESPN.

https://www.espn.com/nba/columns/story?columnist=hollinger_john&id=2850240